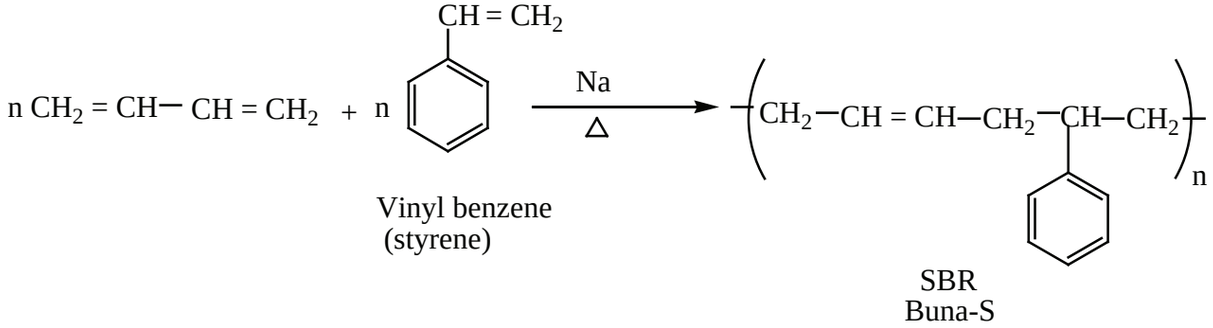


15.4.6 Biodegradable Polymers

The materials that are readily decomposed by microorganisms in the environment are called biodegradable. Natural polymers degrade on their own after certain period of time but the synthetic polymers do not. It leads to serious environmental pollution. One of the solution to this problem is to produce biodegradable polymers which can be broken down by soil micro organism.

Examples:

Polyhydroxy butyrate (PHB)

Poly(3 hydroxy butyrate-co-3-hydroxy valerate) (PHBV)

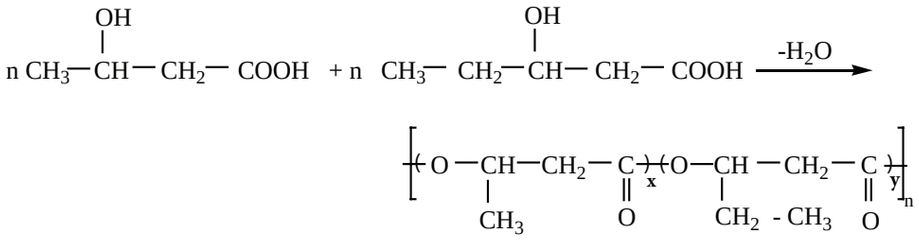
Polyglycolic acid (PGA), Polylactic acid (PLA)

Poly (∈ caprolactone) (PCL)

Biodegradable polymers are used in medical field such as surgical sutures, plasma substitute etc… these polymers are decomposed by enzyme action and are either metabolized or excreted from the body.

Preparation of PHBV

It is the co – polymer of the monomers 3 – hydroxybutanoic acid and 3-hydroxypentanoic acid. In PHBV, the monomer units are joined by ester linkages.



Uses : It is used in ortho paedic devices, and in controlled release of drugs.

Nylon– 2-Nylon -6

It is a co – polymer which contains polyamide linkages. It is obtained by the condensation polymersiation of the monomers, glycine and E - amino caproic acid.

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